

Precision Gauge Theory Simulations and Non-Perturbative Parton Structure at the Meridian High-Energy Facility

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Abstract

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We present high-precision non-perturbative calculations of hadronic matrix elements and parton distribution functions derived from lattice gauge theory simulations at physical quark masses. Using gauge field configurations generated on the Vertex Parallel Supercomputing Cluster with a tree-level improved Symanzik gauge action and $N_f = 2 + 1 + 1$ flavors of highly improved staggered quarks (HISQ), we compute the pion and nucleon vector and axial-vector form factors across momentum transfers up to $Q^2 = 3.0 \text{ GeV}^2$. Continuum extrapolations are performed across three lattice spacings ($a \approx 0.06, 0.09, 0.12 \text{ fm}$) with physical spatial volumes satisfying $m_\pi L > 4.0$. Hadronic matrix elements are renormalized non-perturbatively via the regularized RI'-MOM scheme and converted to the $\overline{\text{MS}}$ scheme at 2 GeV. Our results for the nucleon axial charge $g_A = 1.268(14)$ and the pion decay constant $f_\pi = 130.4(9) \text{ MeV}$ demonstrate sub-percent control over systematic uncertainties, providing key theoretical inputs for ongoing and future scattering experiments at the Meridian Collider Facility.

1 Introduction

Understanding the non-perturbative structure of hadrons from first principles in Quantum Chromodynamics (QCD) remains one of the primary goals of modern subatomic physics. While high-energy hard scattering processes are well described by perturbative QCD expansions in powers of the strong coupling constant $\alpha_s(Q^2)$, low-energy observables and bound-state properties necessitate non-perturbative techniques. Lattice QCD provides a mathematically rigorous, non-perturbative regularization of the gauge theory formulated on a four-dimensional Euclidean spacetime grid.

In recent years, advances in algorithmic design—such as adaptive algebraic multigrid Dirac solvers

and Fourier-accelerated Hybrid Monte Carlo (HMC) algorithms—combined with high-performance computing architectures have enabled lattice simulations to operate directly at physical light-quark masses. This eliminates the reliance on chiral extrapolations from unphysically heavy pion masses, which historically served as a major source of systematic error.

This report summarizes recent progress achieved by the Nexa-Vertex-Synapse Lattice Collaboration in evaluating hadron matrix elements relevant for lepton-hadron scattering experiments conducted at the Meridian High-Energy Collider. Special emphasis is placed on controlling continuum, finite-volume, and excited-state contamination.

2 Theoretical Framework

The Euclidean action for lattice gauge theory consists of the pure gauge action $S_G[U]$ and the fermion action $S_F[\psi, \bar{\psi}, U]$:

$$S[U, \psi, \bar{\psi}] = S_G[U] + S_F[\psi, \bar{\psi}, U]. \quad (1)$$

For the gauge sector, we employ the tree-level improved Symanzik action defined on the hypercubic lattice $\Lambda = \{x = (x_1, x_2, x_3, x_4) \mid x_\mu/a \in \mathbb{Z}\}$:

$$S_G[U] = \frac{\beta}{3} \sum_{x, \mu < \nu} \left(c_0 \text{Re Tr} [1 - P_{\mu\nu}(x)] + c_1 \text{Re Tr} [1 - R_{\mu\nu}(x)] \right), \quad (2)$$

where $P_{\mu\nu}(x)$ denotes the standard 1×1 plaquette loop, $R_{\mu\nu}(x)$ represents the rectangular 1×2 loop, $\beta = 6/g^2$ is the lattice gauge coupling, and the normalization condition requires $c_0 + 8c_1 = 1$ with $c_1 = -1/12$.

The fermion action uses the $N_f = 2 + 1 + 1$ Highly Improved Staggered Quark (HISQ) formulation for sea quarks and the Wilson-clover action for valence